

Unless otherwise stated, each problem is worth 4 points. Show all of your work.

Evaluate the following limits.

1. $\lim_{(x,y) \rightarrow (1,\pi)} (3x + \sin xy)$

2. $\lim_{(x,y) \rightarrow (1,-1)} \frac{y^3 + y^2x}{y + x}$

Use the Two-Path Test to prove that the following limit do not exist.

3. $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - 4y^3}{x^2 + 2y^3}$

Find the indicated partial derivative of the following functions.

4. Find g_x and g_y for $g(x, y) = (x^2 + 2)e^y$.

5. [8 pts] Find the four second partial derivatives of $h(x, y) = x^2 \cos 2y$.

Use the Chain Rule to find the following derivative. When feasible express your answer in terms of the independent variable.

6. [6 pts] dz/dt where $z = x^3y - x$, $x = t^2$, and $y = 2t^{-1}$.

Compute the gradient of the following function and evaluate it at the given point P .

7. $f(x, y) = x^2 + y^2$; $P(-1, 2)$

Consider the following functions and points P .

- a. Find the unit vectors that give the direction of steepest ascent and steepest descent at P .
 - b. Find a vector that points in a direction of no change in the function at P .
8. [6 pts] $f(x, y) = x^2 + y^2$; $P(-1, 2)$

Use the Second Derivative Test to determine (if possible) whether each critical point of the given function is a local maximum, local minimum, or saddle point.

9. [8 pts] $f(x, y) = 4 + x^3 + y^3 - 3xy$ with critical points $(0, 0)$ and $(1, 1)$.

Find all of the critical points (there are two) of the following function.

10. $f(x, y) = \frac{x^3}{3} - \frac{y^3}{3} + 3xy$